

Airbnb

You are a Product Analyst on the Airbnb Stays team focusing on host communication patterns....

Question 1: What is the average response time for hosts during April 2024, and how does it vary across different property types?

```
SELECT property_type, AVG(response_time_minutes) AS average_response_time
FROM fct_host_communications
INNER JOIN dim_properties USING (property_id)
WHERE TO_CHAR(communication_date, 'YYYY-MM') = '2024-04'
GROUP BY property_type
```



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Question 2: Identify the top 5 hosts with the fastest average response times in April 2024. How does their response time compare to the overall average response time for that month?

```
WITH each_host AS (  
    SELECT host_id, AVG(response_time_minutes) AS average_response_time  
    FROM fct_host_communications  
    WHERE TO_CHAR(communication_date, 'YYYY-MM') = '2024-04'  
    GROUP BY host_id  
)  
overall_avg AS (  
    SELECT AVG(response_time_minutes) AS overall_average_response_time  
    FROM fct_host_communications  
    WHERE TO_CHAR(communication_date, 'YYYY-MM') = '2024-04'  
)  
  
SELECT  
    eh.host_id,  
    eh.average_response_time,  
    oa.overall_average_response_time,  
    eh.average_response_time - oa.overall_average_response_time  
AS difference  
FROM each_host eh  
CROSS JOIN overall_avg oa  
ORDER BY 2  
LIMIT 5
```



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Question 3: For the second quarter of 2024, calculate the average host response times and booking conversion rates for each property. This will help us understand if a quicker response time leads to a higher booking conversion rate.

```
WITH response AS(
    SELECT property_id, AVG(response_time_minutes) AS average_host_response_times
    FROM fct_host_communications
    WHERE TO_CHAR(communication_date, 'YYYY-MM') IN ('2024-04', '2024-05', '2024-06')
    GROUP BY property_id),

    booking AS(
    SELECT property_id,
        SUM(CASE WHEN booking_status = 'confirmed' THEN 1 ELSE 0 END)*100.0 / COUNT(*) AS conversion_rate
    FROM fct_bookings
    WHERE TO_CHAR(booking_date, 'YYYY-MM') IN ('2024-04', '2024-05', '2024-06')
    GROUP BY property_id)

SELECT *,
    average_host_response_times,
    conversion_rate
FROM dim_properties AS d
INNER JOIN response AS r
    ON d.property_id = r.property_id
INNER JOIN booking AS b
    ON d.property_id = b.property_id
```

